

Spokes: Reproducible Board Reporting for The Northwest Hub

M5 Final Write-Up

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1 Title, Team, and Affiliations

Project title: Spokes: Reproducible Board Reporting for The Northwest Hub

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Affiliations:

- Data Science Capstone, Willamette University
- Community partner: The Northwest Hub

Repository: <https://github.com/thenorthwesthub/spokes> (The Northwest Hub and Contributors 2026)

2 Abstract

The Northwest Hub needs board-ready reporting without recurring spreadsheet reconciliation across Lightspeed, QuickBooks, and program/grant workflows. Spokes implements a reproducible reporting pipeline using scheduled Python ETL in GitHub Actions, a Neon Postgres warehouse, and Grafana dashboards hosted on Fly.io ([The Northwest Hub and Contributors 2026](#); [Neon 2026](#); [Grafana Labs 2026](#)).

This final write-up freezes claims to what was defended at poster session: the Lightspeed reporting slice is operational and reproducible, while QBO unattended OAuth hardening and program-impact metric development remain partial. The primary research question is answered for the Lightspeed scope: leadership can use live dashboards instead of manual POS spreadsheet reconciliation.

The secondary research question is answered with descriptive trend analysis: directional-change comparisons show sales and ticket count rising faster than average basket size, indicating volume-led growth. All conclusions are bounded to validated data windows and known source limitations.

3 Introduction: Problem and Research Questions

The Northwest Hub tracks operations and finance across multiple systems, including Lightspeed R-Series POS and QuickBooks Online. Before this project, recurring leadership reporting depended on manual exports and spreadsheet reconciliation. This introduced delay, inconsistencies in metric definitions, and reproducibility risk.

The capstone objective is both technical and operational: design a low-cost, maintainable reporting workflow that produces board-usable metrics on a predictable cadence with explicit quality boundaries.

Primary research question

How can The Northwest Hub maintain a reproducible, low-maintenance reporting workflow that consolidates key operations and financial indicators for board decision support?

Secondary research questions

1. What does trailing-window KPI behavior (sales, tickets, average basket) show relative to prior-window and same-period-last-year baselines?
2. Are observed changes primarily volume-driven, basket-driven, or mixed, and can this be evaluated using directional-change analysis?
3. What quality boundaries must be attached to those findings?

4 Impact and Stakeholders

Spokes is designed for stakeholders who currently spend time assembling monthly reporting packets from multiple systems.

Named stakeholder groups

- Executive leadership.
- Board members and advisors.
- Staff who manage their source systems and reports.

Intended impact pathways

1. Reduce manual reconciliation burden for recurring Lightspeed KPI reporting.
2. Standardize metric definitions in SQL/query logic rather than ad-hoc spreadsheet formulas.
3. Keep track and improve decision timeliness by making trend and freshness signals visible in one dashboard surface.

5 Architecture and Source Integration

1. **Pipeline architecture:** GitHub Actions + Neon + Grafana replaced a heavier orchestration path to reduce operational overhead while preserving reproducibility ([The Northwest Hub and Contributors 2026](#); [Neon 2026](#); [Grafana Labs 2026](#)).
2. **Source integration constraints:** both QBO and Lightspeed rely on OAuth-based API access, with different operational failure modes and token lifecycle behavior ([Intuit Developer 2026](#); [Lightspeed](#)

[Developers 2026](#); [Denniss, Bradley, and Sakimura 2017](#)).

This places the project in applied analytics engineering: reproducible descriptive reporting first, then expanded source completeness.

6 Data and Data Engineering

6.1 Sources and implemented schema

The warehouse in this milestone is a reporting mart centered on `financial_balances`, with supporting tables `vendor_liabilities` and `program_impact`. The schema is defined in `src/fin_balance.py`.

- `financial_balances` unique grain: `(as_of_date, source_system, line_item)`
- `vendor_liabilities`: payable snapshots by vendor/aging category
- `program_impact`: mixed-type program metrics currently stored as text

The reporting view layer is built in `src/transform_data.py` as three SQL views:

- `view_grafana_monthly_pnl`
- `view_grafana_weekly_ops`
- `view_grafana_vendor_trends`

6.2 Source status at M5 close

Source	Status in M5	Evidence role
Lightspeed R-Series POS	Operational scheduled hourly sync (<code>lightspeed_sync.yml</code>)	Primary evidence basis for KPI findings
QuickBooks Online	Partial (POC pull exists; unattended hardening pending)	Secondary/partial evidence only
Google Sheets source rows	Ingestion path exists via <code>google_sheet_rows</code> ; not core to defended KPI claims	Supporting pipeline capability

6.3 Engineering controls used

1. Token lifecycle handling and persistence for Lightspeed OAuth refresh flow.
2. UPSERT write semantics for Lightspeed KPI snapshots to prevent duplicate growth from retries.
3. Backfill and missed-day repair routines to restore date coverage continuity.
4. Explicit freshness and coverage checks used in dashboard interpretation.

Freshness signals in the Data Health Checks dashboard are computed from warehouse write timestamps (`updated_at`) rather than date-only snapshot keys (`as_of_date`), so the lag indicator represents true hourly load recency.

7 Ethics and Responsible Use

This project handles private financial and operational information for a community-serving organization. The main ethical focus is data stewardship and claim discipline.

Data handling and confidentiality are governed by a signed NDA with The Northwest Hub; that agreement is managed outside this repository and is therefore not included as a repo artifact. Within the repository, governance controls are evidenced through claim-boundary discipline, source-status labeling, and reproducible pipeline documentation.

Considerations regarding data use and interpretation are framed as responsible reporting and decision support. Potential community harm from misinterpretation is mitigated by explicit claim boundaries, transparent source limitations, and visible freshness/coverage checks in dashboards.

Controls applied

1. Public-facing artifacts do not expose raw sensitive records.
2. Credentials are handled through environment variables and secret stores, not committed in source.
3. Findings are bounded to validated windows and explicitly label partial source coverage.

Interpretation boundaries

1. Results are descriptive, not causal.
2. Trend changes are decision signals, not proof of intervention impact.
3. Historical gaps and source incompleteness are reported as limitations, not silently ignored.

8 Methods and Evaluation

Methodology combines engineering reproducibility checks with descriptive trend analysis.

8.1 Design

1. Build a reproducible source-to-dashboard pipeline for recurring board reporting.
2. Quantify KPI direction and magnitude using trailing-window comparisons.
3. Attach confidence boundaries through freshness and coverage checks.

8.2 Analytical procedure

1. Pull completed Lightspeed sales and transform into KPI rows (`POS Sales` , `Ticket Count` , `Avg Basket Size`).

2. Load KPI snapshots into `financial_balances`.
3. Build weekly/monthly trend and rollup views.
4. Compute directional changes for trailing windows against:
 - prior 90-day window
 - same period last year (SPLY)

Directional change is calculated as:

8.3 Evaluation criteria

Engineering readiness

1. Core scripts run successfully for schema, pull, and transform layers.
2. Dashboard panels render from warehouse-backed queries.
3. Rebuild instructions are sufficient for another analyst to reproduce outputs.

Analytical evidence quality

1. Claims trace to explicit query logic and warehouse records.
2. Every comparative claim states its reference window.
3. Limitations are attached when source coverage is partial.

8.4 Scope boundary

No predictive model training is used. Train/validation/holdout paradigms are not applicable to this milestone.

9 Results and Visualizations

9.1 Engineering result: Lightspeed reconciliation workflow replaced

For the Lightspeed scope, the project meets its core operational goal. Leadership can review recurring KPIs from the live dashboard instead of manually reconciling POS exports.

Evidence includes:

1. Scheduled sync behavior with reproducible load/transform scripts.
2. Dashboard panels for MTD sales, ticket count, average basket, and confidence checks.
3. Warehouse and dashboard artifacts versioned in repository paths.

9.2 Analytical result: growth is primarily volume-led

Trailing-window directional-change results defended at poster session:

Metric	vs Prior 90D	vs SPLY
Sales	+46.9%	+5.4%
Ticket Count	+37.2%	+7.8%
Average Basket	+7.1%	-2.2%

Interpretation: sales and ticket growth outpace basket growth, indicating traffic/transaction volume is the dominant driver in the evaluated period.

9.3 Visual evidence

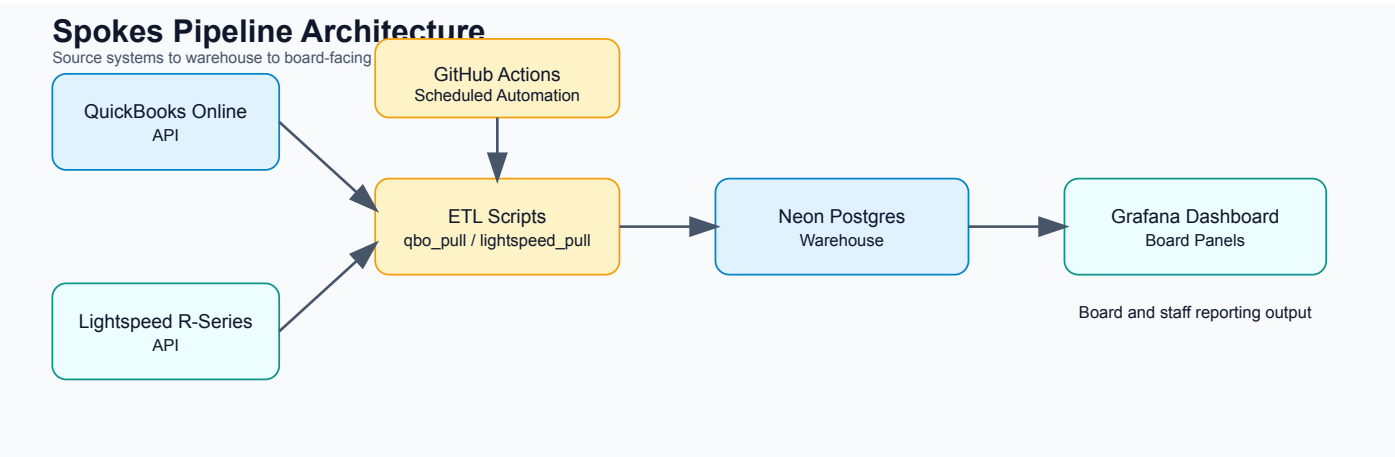


Figure 1: Pipeline architecture used in the defended poster and repository docs.

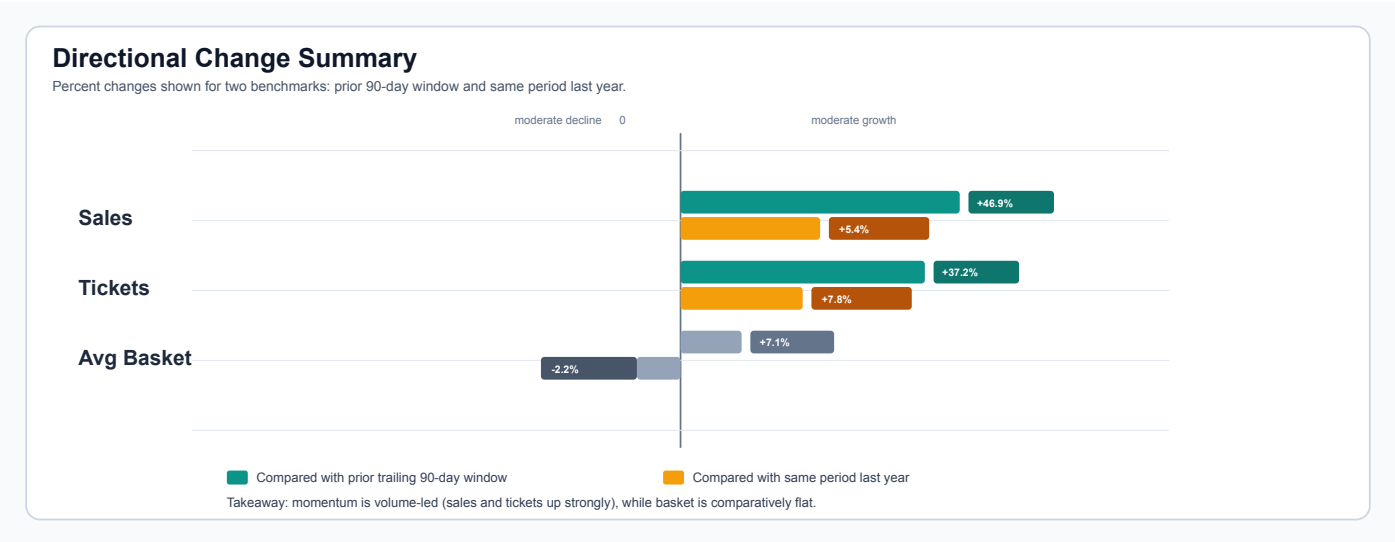


Figure 2: Trailing-window directional-change summary (obfuscated values for public sharing).

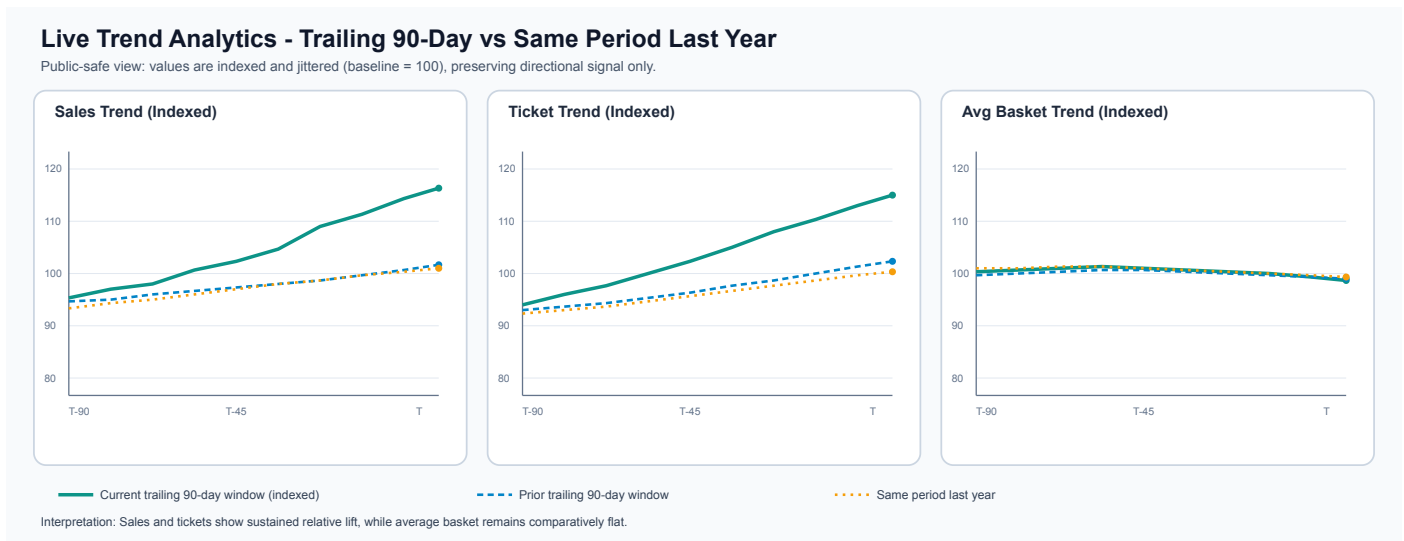


Figure 3: Live trend panel pattern used for interpretation (indexed/obfuscated).

10 Discussion, Limitations, and Future Work

10.1 Discussion

The strongest contribution is not only trend insight but process reliability: metrics now flow through a repeatable, inspectable path with explicit confidence signals. That is a practical governance gain for board reporting.

10.2 Limitations

1. Source maturity is uneven: Lightspeed is operational; QBO unattended flow remains incomplete.
2. Program-impact table values are text-typed, limiting numeric analytics without additional typing/validation.
3. This work supports descriptive decision intelligence, not causal attribution.

10.3 Future work

1. Complete QBO unattended OAuth hardening and remove manual token dependency.
2. Harden program-impact schema typing and validation rules for numeric indicators.
3. Expand dashboard confidence checks to source-specific quality flags and alerting.
4. Introduce optional dimensional decomposition if broader cross-domain analytics requires it.

11 Conclusions and Recommendations

Spokes achieved a production-relevant milestone for the Lightspeed reporting slice: recurring board KPI review can be performed from reproducible warehouse-backed dashboards rather than manual spreadsheet reconciliation.

The current analytical finding is actionable: growth is volume-led more than basket-led in the evaluated windows. Near-term operations should prioritize traffic and repeat-visit levers while testing basket-lift tactics in controlled cycles.

Recommendations:

1. Preserve claim boundaries: treat Lightspeed findings as production-grade; treat QBO/program claims as partial until hardening is complete.
2. Keep freshness and coverage panels visible in executive views to anchor decision confidence.
3. Sequence phase-two work around QBO unattended reliability and typed program metrics before expanding interpretation scope.
4. Keep runbook and dashboard docs synchronized using the documentation precedence map ([docs/documentation/source-of-truth.md](#)) so handoff instructions stay aligned with live behavior.

12 Reproducibility and References

12.1 Reproduction workflow

```
# 1) From a local clone of thenorthwesthub/spokes
cd spokes
pip install -r requirements.txt
cp .env.example .env
python src/test_db.py
python src/fin_balance.py
python src/transform_data.py

# 2) Verify pull output first (dry run), then load only when approved
python src/lightspeed_pull.py
python src/lightspeed_pull.py --load

# 3) Use the async handoff runbook checklist for verification evidence
# docs/runbooks/async-handoff.md

# 4) Render this writeup from a local clone of socio-economic
cd ../socio-economic
quarto render deliverables/M5-final/writeup.qmd
```

Notes:

1. [src/load_initial_data.py](#) is non-idempotent and can duplicate rows on re-run.
2. QBO POC currently depends on short-lived access tokens unless further hardening is applied.

12.2 Key repository paths

1. `src/fin_balance.py`
2. `src/lightspeed_pull.py`
3. `src/qbo_pull.py`
4. `src/google_sheets_pull.py`
5. `src/transform_data.py`
6. `.github/workflows/lightspeed_sync.yml`
7. `.github/workflows/google_sheets_sync.yml`
8. `docs/runbooks/lightspeed-api.md`
9. `docs/runbooks/async-handoff.md`
10. `docs/executive-analytics-grafana/panel-queries.sql`
11. `grafana_fly/dashboards/data-health.json`
12. `docs/documentation/source-of-truth.md`

12.3 References

Primary references and API/documentation sources are cited in-text and listed in the bibliography ([The Northwest Hub and Contributors 2026](#); [Intuit Developer 2026](#); [Lightspeed Developers 2026](#); [Grafana Labs 2026](#); [Neon 2026](#); [Denniss, Bradley, and Sakimura 2017](#)).

References

- Denniss, W., J. Bradley, and N. Sakimura. 2017. "OAuth 2.0 for Native Apps (RFC 8252)." <https://www.rfc-editor.org/rfc/rfc8252>.
- Grafana Labs. 2026. "Grafana Documentation." <https://grafana.com/docs/>.
- Intuit Developer. 2026. "QuickBooks Online API Documentation." <https://developer.intuit.com/app/developer/qbo/docs/api/accounting/all-entities>.
- Lightspeed Developers. 2026. "Lightspeed Retail (r-Series) API Documentation." <https://developers.light speedhq.com/retail/endpoints/>.
- Neon. 2026. "Neon Documentation." <https://neon.tech/docs>.
- The Northwest Hub and Contributors. 2026. "Spokes Repository." <https://github.com/thenorthwesthub/spokes>.